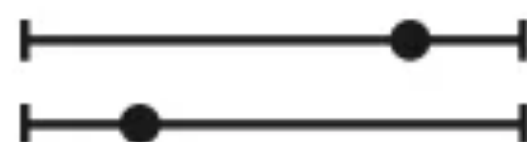


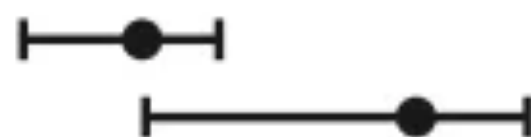


➔ **Magnitude Channels: Ordered Attributes**

Position on common scale



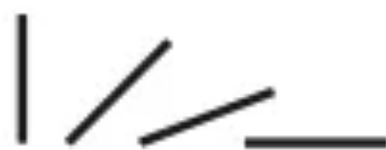
Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



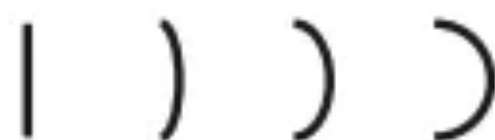
Color luminance



Color saturation



Curvature



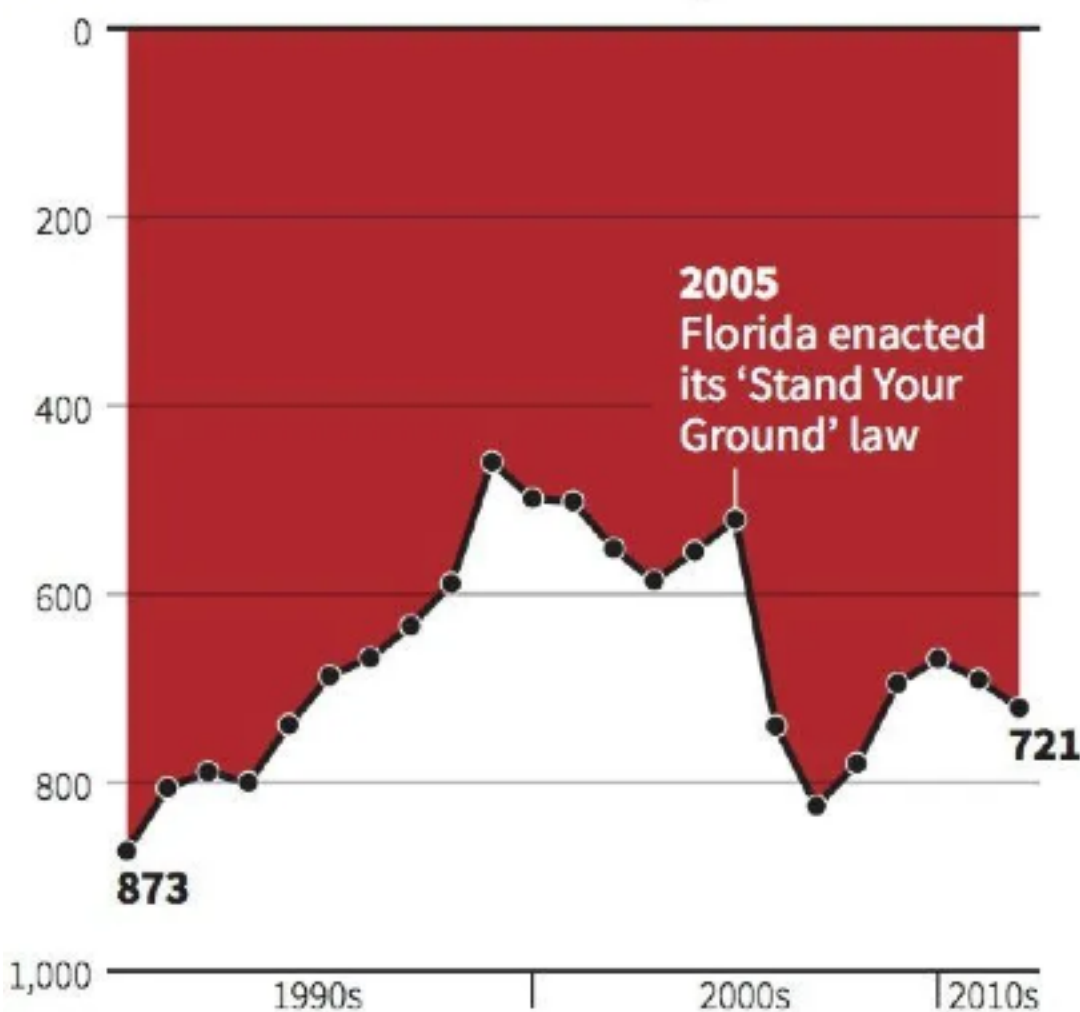
Volume (3D size)



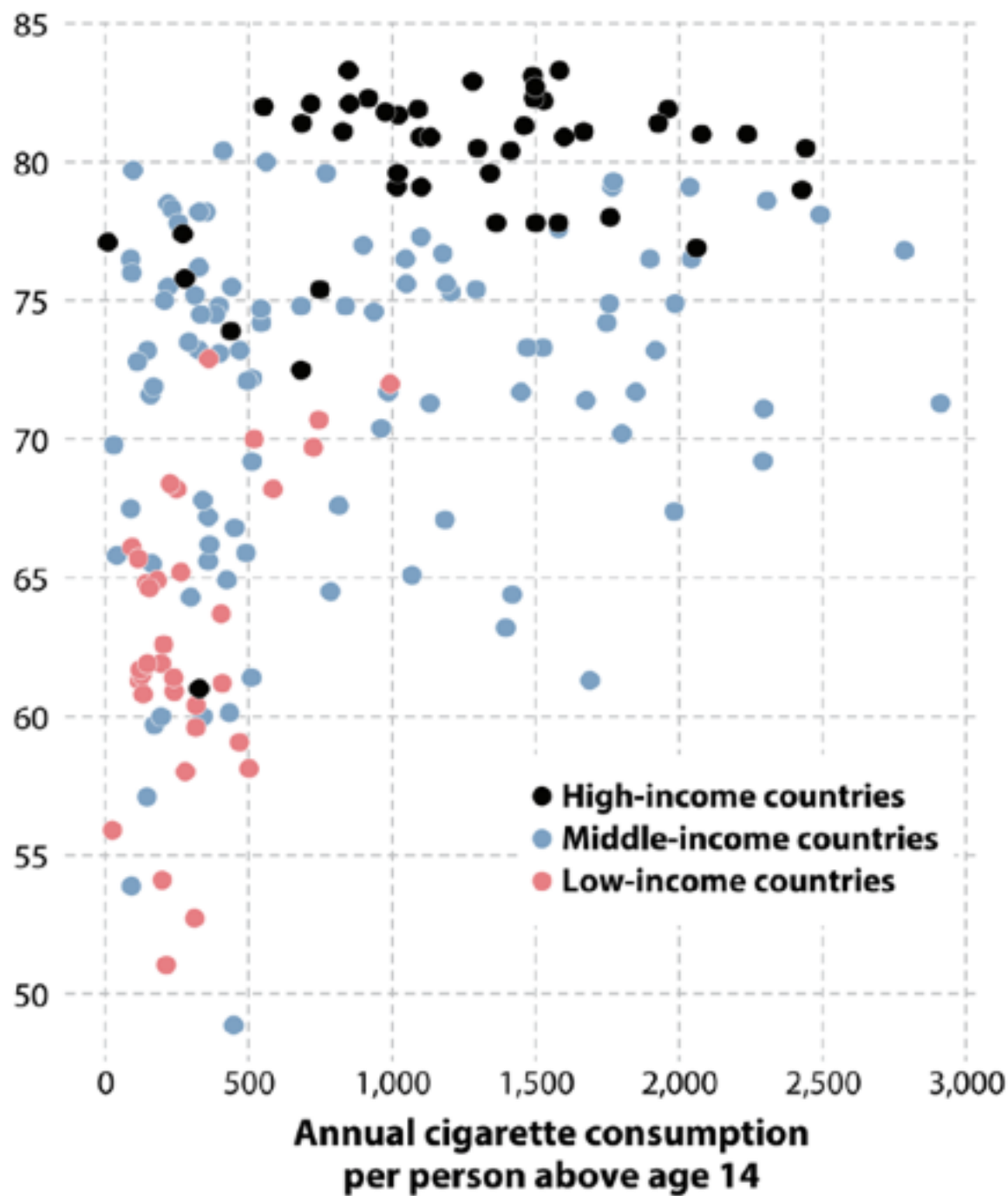
Tamara Munzner, Visualization Analysis and Design (2014)

Gun deaths in Florida

Number of murders committed using firearms



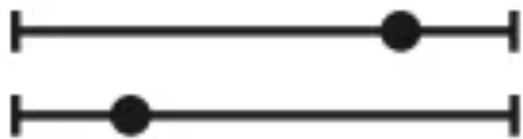
Source: Florida Department of Law Enforcement



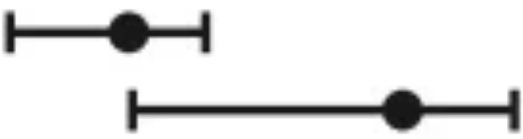


➔ **Magnitude Channels: Ordered Attributes**

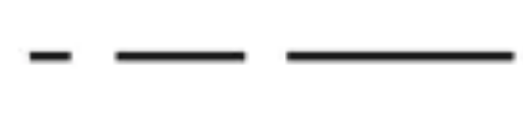
Position on common scale



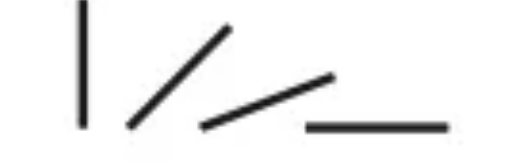
Position on unaligned scale



Length (1D size)



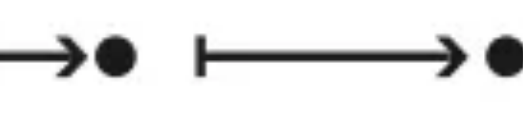
Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature

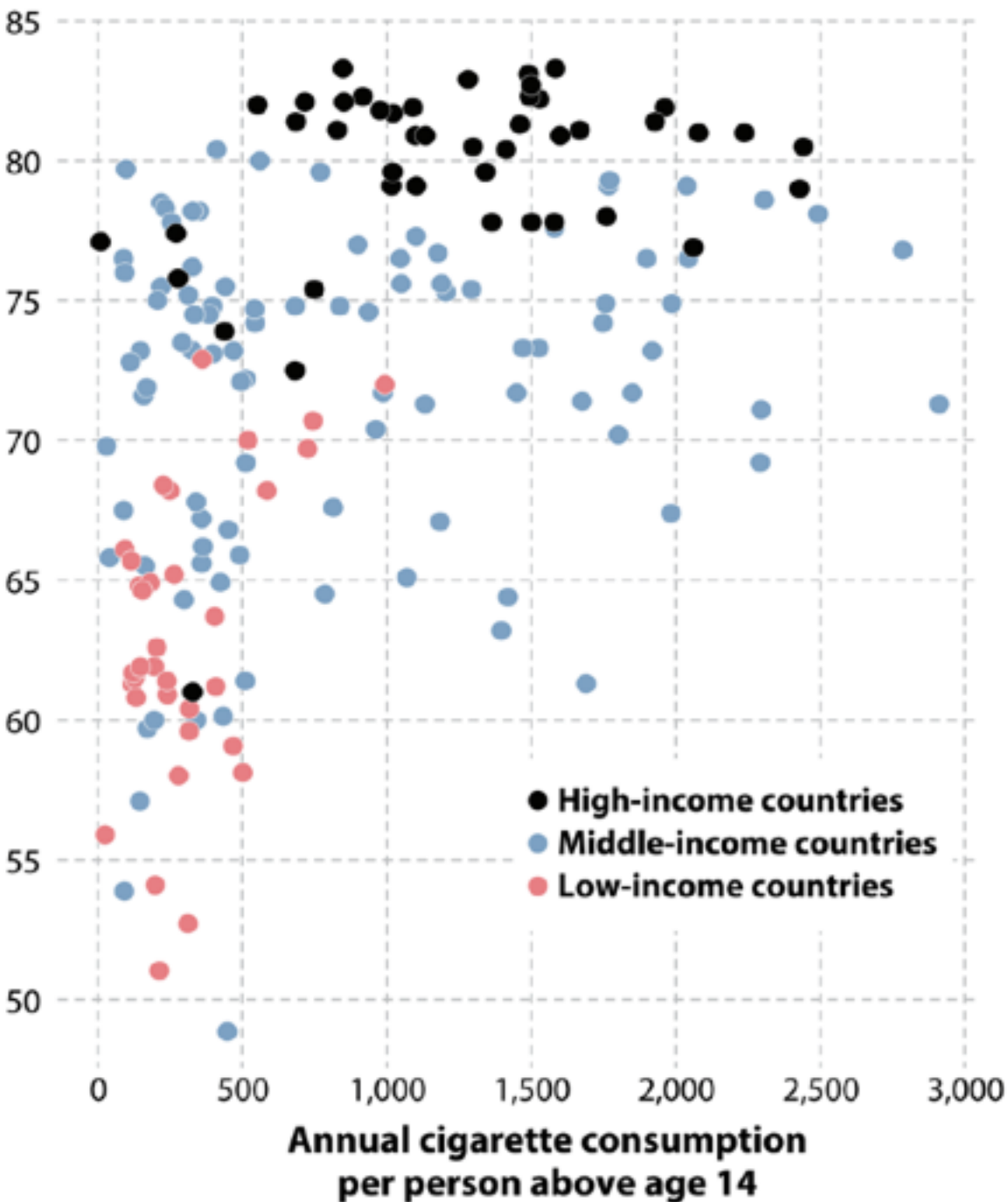
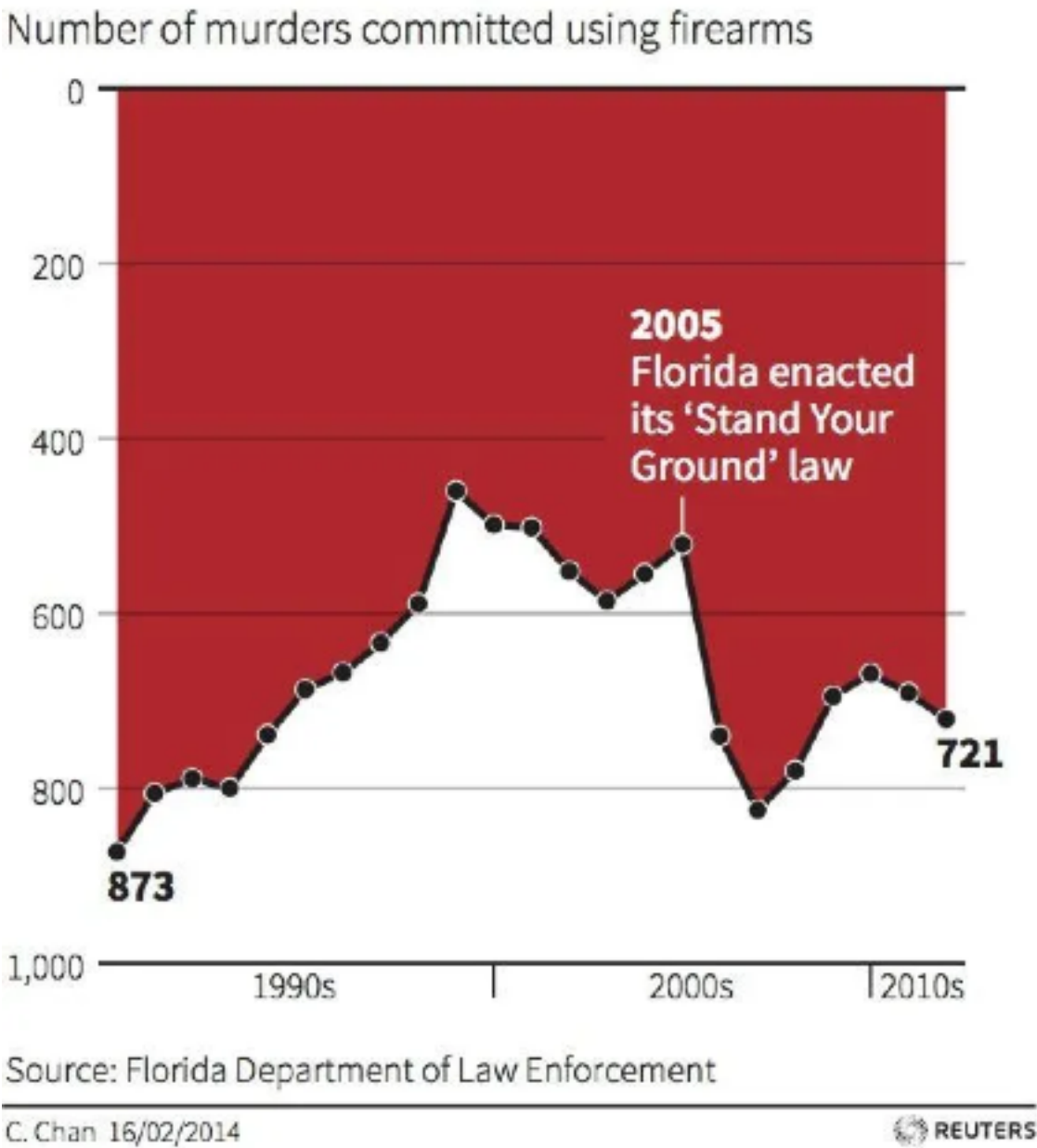


Volume (3D size)



Data  Visual
Expressiveness Effectiveness

Gun deaths in Florida



Week 3: Perception and Cognition

CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

February 16, 2026

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>